18.6.2 Diffusion

Diffusion is the movement of particles from an area of **higher concentration to lower concentration**. The overall effect is to equalize concentration throughout the medium.

Examples for diffusion include, perfume filling a whole room and the movement of small molecules across a cell membrane. One of the simplest demonstrations of diffusion is adding a drop of ink to water.

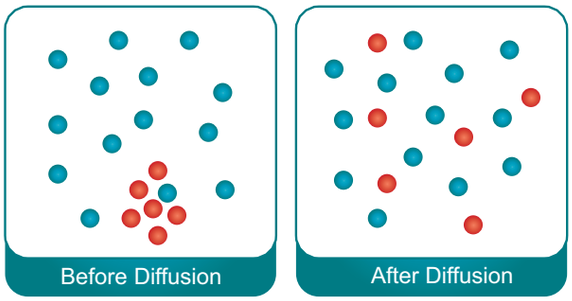


Figure 18.9 Diffusion of gases

What will happen when an incense stick is lit up in a room? How do we feel? The fragrance spreads over the entire room. The movement of molecules or ions is from a region of higher concentration to region of lower concentration. You can smell incense stick after lighting because the smoke diffuses in the air and makes its way to your nose. Let us think of the following. How does the smell spread in the entire room? Does the smell spread uniformly in the entire room? Can you give any other examples?

There are other processes in which substances move in liquid medium. For an example when a tea pack is immersed in a cup of hot water the tea powder particles disperse in to water by diffusion.



Figure 18.10 Diffusion through liquid medium

DO YOU KNOW?

The mixing of foodstuffs and digestive juices in the gut occurs by diffusion. Exchange of respiratory gases (Oxygen and Carbon dioxide) between blood and tissue fluids between tissue fluid and cells also occurs by diffusion.

18.6.3 Osmosis

Osmosis is the movement of solvent particles across a semipermeable membrane from a dilute solution into a concentrated solution. The solvent moves to dilute the concentrated solution and equalize the concentration on both sides of the membrane.

The movement of liquids in and out of cells is dependent on the concentration of the solution surrounding it. There are three types of situations in which this could vary.

Isotonic

Here the concentration of external and internal solution of the organism are the same.

Hypotonic

Here the external solution concentration is less compared to the concentration of the inner solution of an organism. In this case water will rush into the organism.

Hypertonic

Here the external solution concentration is greater than the concentration of the inner solution of an organism. In this case the water will rush out of the organism.

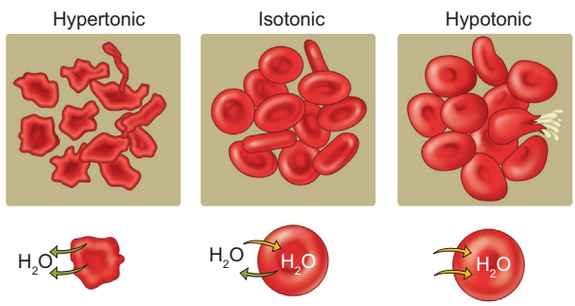


Figure 18.11 Osmosis in red blood cells

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